nano3DX

High-resolution 3D X-ray microscope

Submicron X-ray CT system

Rigaku
POWERING NEW PERSPECTIVES

nano3DX

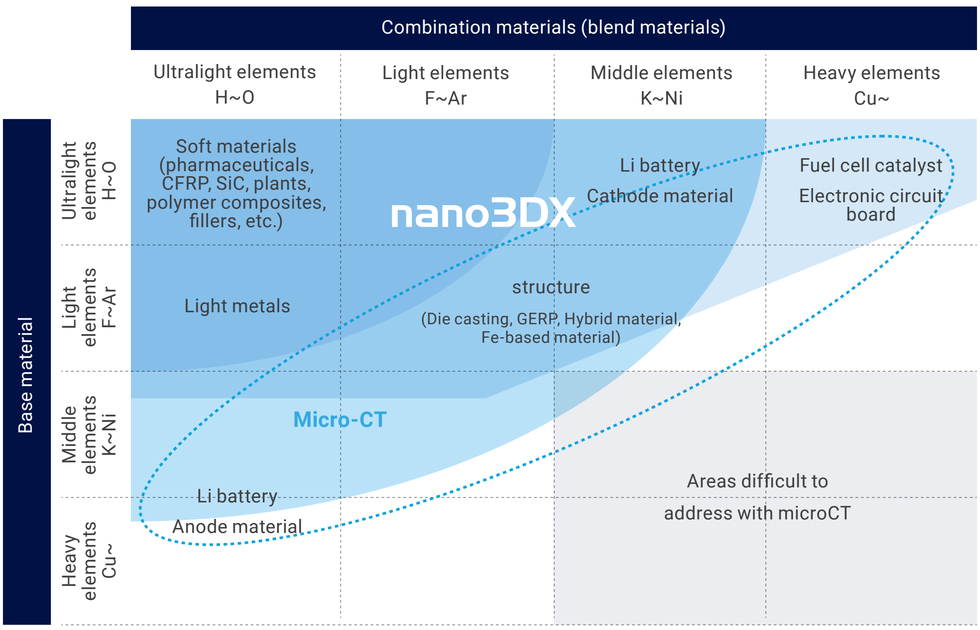
True submicron CT scanner with contrast-enhancing X-ray anode.

Easy CT imaging and CT reconstruction.

High-resolution 3D X-ray microscope.

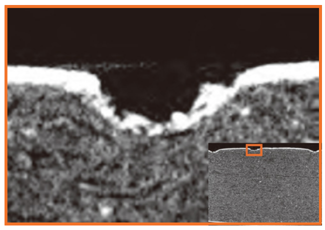
Supports a wide range of applications

The nano3DX is a true submicron-resolution X-ray microscope (XRM). The parallel beam geometry combined with an ultrabright 1200 W rotating anode source enhances the contrast of soft materials, which are normlly difficult to image using high-energy X-ray sources. Applications include pharmaceuticals, polymers, composites, inorganic materials, light metals, and life science specimens, as well as small electronic components.

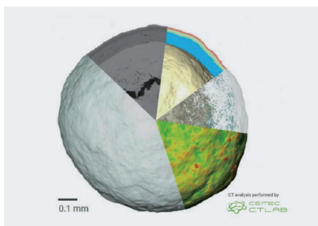


Pharma formulation

Tablet coating

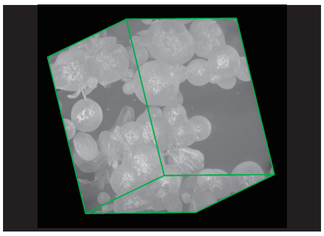


Granules

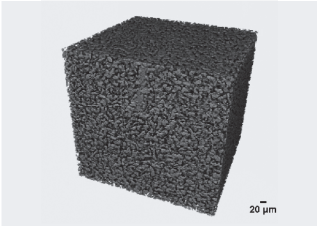


Polymer • composite material • inorganic material

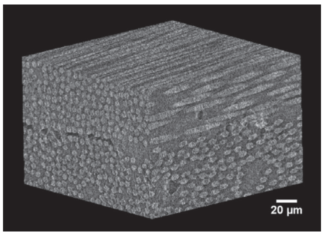
Polymer blends



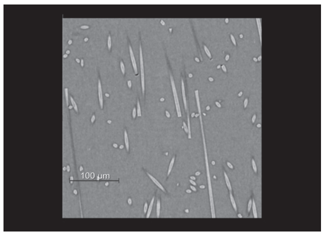
Filler composite



CFRP

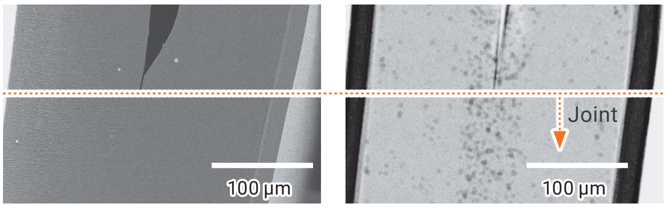


CFRP



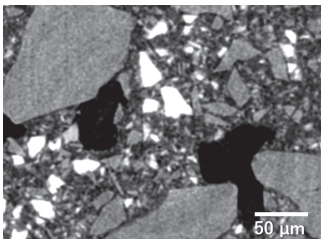
Sample provided by: Professor Tashiro, Toyota Technological Institute

Cross-sectional SEM image of a nylon plastic bag joint (left) X-ray transmission image taken from the cross-sectional direction (right)

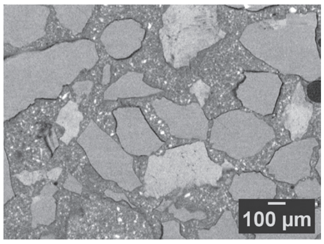


Sample provided by: Toray Research Center, Inc.

Ceramic composites

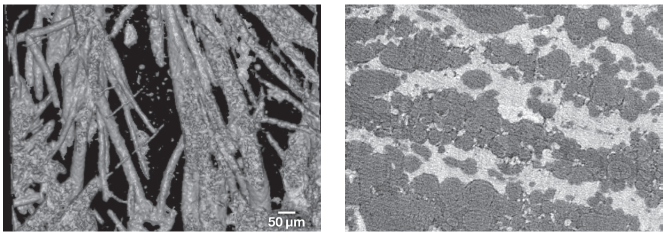


Inorganic material



Light metals

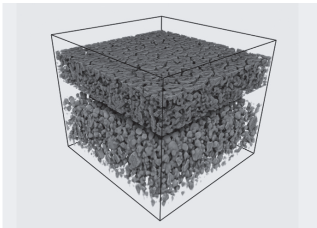
High thermal conductivity Al-C composite



Sample provided by: Single-Walled Carbon Nanotube Fusion New Materials Research and Development Organization / Osaka Prefectural Technology Research Institute / Hokkaido University

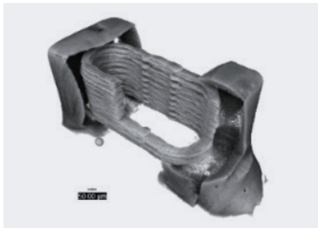
Batteries/electronic materials

Battery materials



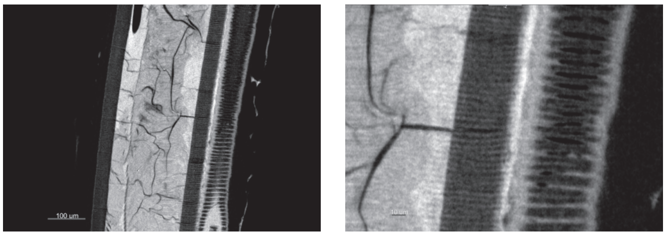
Sample provided by: LIBTEC

Electronic components



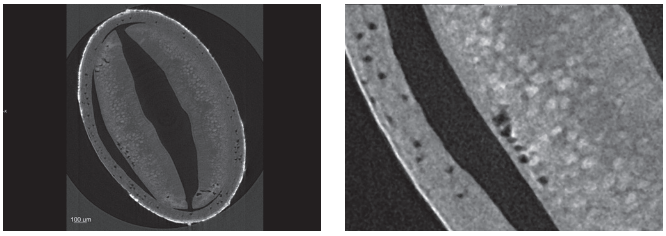
Life Sciences

Butterfly antennae

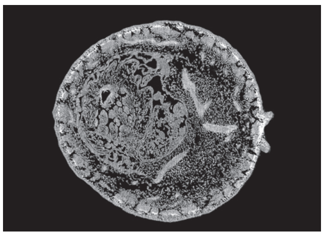


Sample: Courtesy of Prof. K. G. Kornev, Clemson University, USA

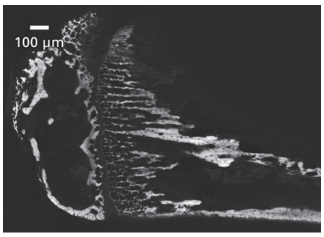
Sesame seeds



Nandina berries

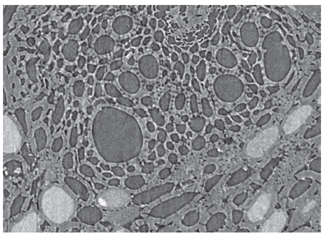


Bone

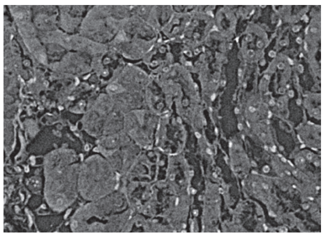


Sample provided by Professor Matsuo, Keio University

Plant



Organs (kidneys)



High 2D/3D spatial resolution imaging with true submicron resolution

High 2D/3D spatial resolution

- A high-resolution X-ray detector supports voxel sizes down to 188 nm
- Parallel beam geometry combined with lens-based magnification is immune to blurring due to X-ray focus size and drifting
- A high-precision 5-axis stage enables accurate, repeatable CT acquisition for demanding metrology and microstructure studies

High contrast (density resolution)

- Choose the characteristic X-ray target to match your sample and imaging goals
- The X-ray anode can be selected from Cr (5.4 keV), Cu (8 keV), or Mo (17 keV) for low energy, characteristic radiation to maximize the contrast for the given sample material and size
- Optional phase retrieval further enhances soft-tissue and low-absorption contrast for challenging samples

High-speed imaging

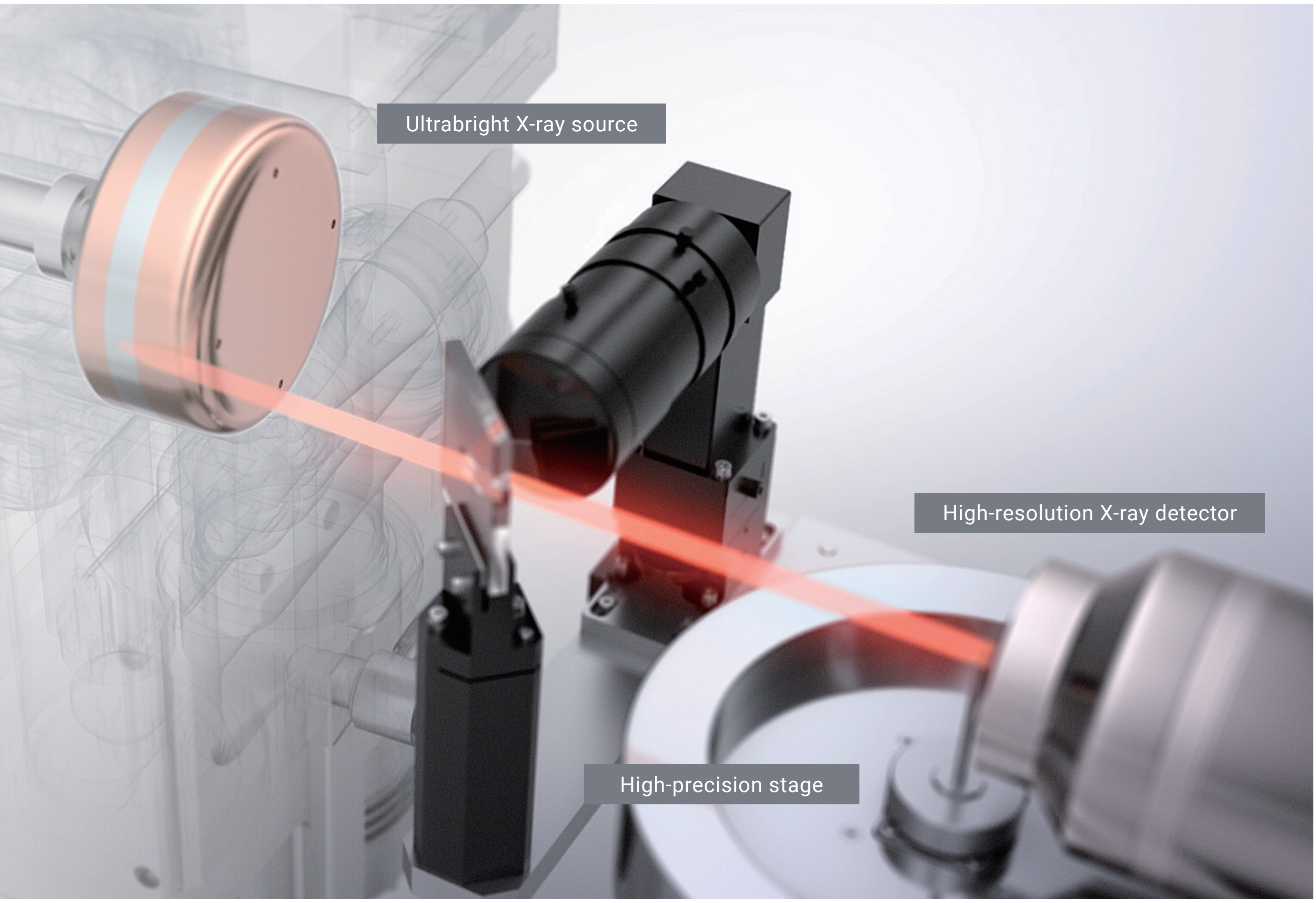
- The ultrabright 1200 W rotating anode X-ray generator paired with a high-resolution detector enables rapid CT data collection, capturing micron-scale structural information in minutes

Quasi-parallel beam optical system

The system uses a parallel beam geometry similar to that used at synchrotron facilities and an optical lens for magnification. This geometry is immune to blurring due to X-ray focus size and drifting and is suitable for high-resolution (submicron) X-ray CT measurements.

High-brightness X-ray source 1200 W

Our proprietary high-intensity X-ray generator delivers up to 1200 W, providing roughly 100X higher intensity than typical microfocus sources used for high-resolution imaging (generally a few watts). The system features easy selection of low-energy characteristic X-rays from dual-source targets (Cr, Cu, Mo, W). The X-ray source is built on technologies refined over many years to provide excellent source stability during extended, continuous operation.



High-resolution X-ray detector

0.188~3.00 μm/voxel

The system features a high-resolution X-ray detector supporting voxel sizes down to 0.188 μm, with a field of view 2X larger than conventional models. Paired with an ultrabright source providing characteristic X-rays, the system provides low-noise images with strong contrast.

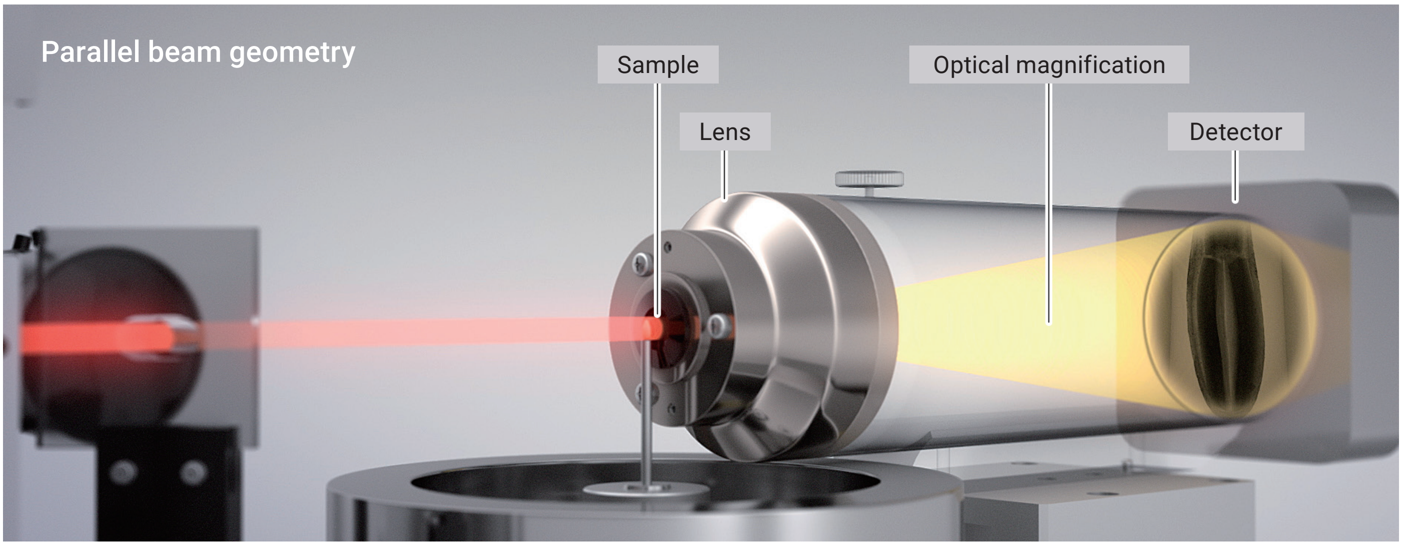
High-precision stage

Sub-micron rotational stability (≤ 1 μm runout)

The system includes a high-precision stage with core runout of 1 μm or less. It leverages the same high-accuracy rotary axis technology used in Rigaku's flagship X-ray diffraction instruments, providing 0.0001° angular resolution to support high-resolution image acquisition.

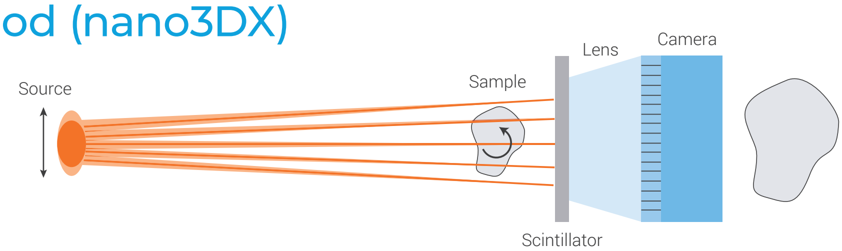
Easy CT imaging and reconstruction

- Equipped with a high-resolution X-ray detector (0.188 μm/voxel at maximum magnification)
- Parallel beam geometry combined with lens magnification preserves resolution and minimizes blurring due to focal spot size effects
- High-precision CT imaging using a high-precision 5-axis sample stage (with core runout of 1 μm or less)



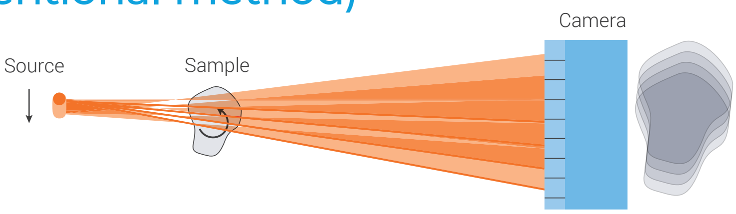
Quasi-parallel beam method (nano3DX)

Because the system uses parallel beam geometry for high-magnification imaging, the optical system is less sensitive to X-ray source fluctuations.



Cone beam method (conventional method)

With cone beam systems, even small fluctuations in the X-ray source or sample motion are amplified, which can lead to image blur.



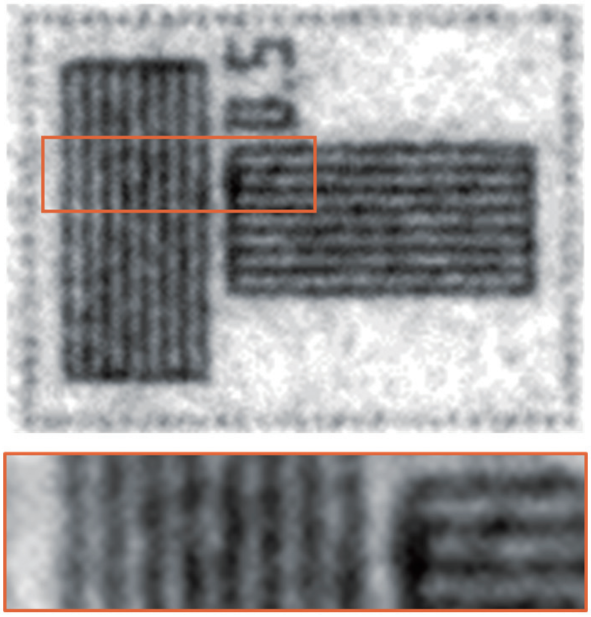
Lens list

Lens	FOV mm	Voxel size μm
L0270	φ0.90×0.75	0.188
L0540	φ1.80×1.50	0.376
L1080	φ3.60×3.00	0.752
L2160	φ7.20×6.00	1.50
L4320	φ14.4×12.0	3.00

※ All models except L0270 are optional
※ Easily replaceable depending on the sample size

- Five types of lenses are available, covering apertures from φ0.90 to φ14.4 mm
- Select the lens appropriate for your imaging goals, whether you need a small field of view with higher resolution or a larger field of view at lower magnification
- The field of view can be expanded by approximately 1.6 times using the optional offset scan mode

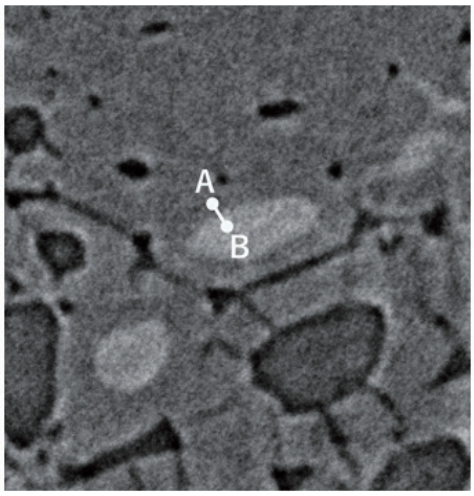
High resolution (2D)

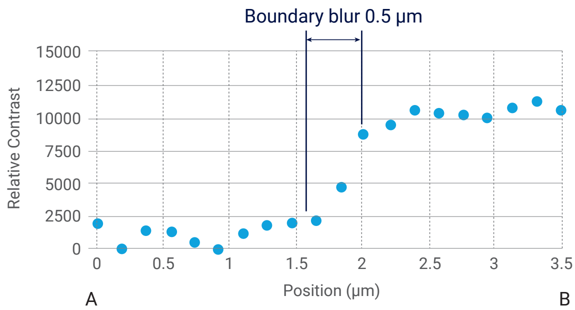


JIMA RT RC-04 (1 line = 0.5 μm)

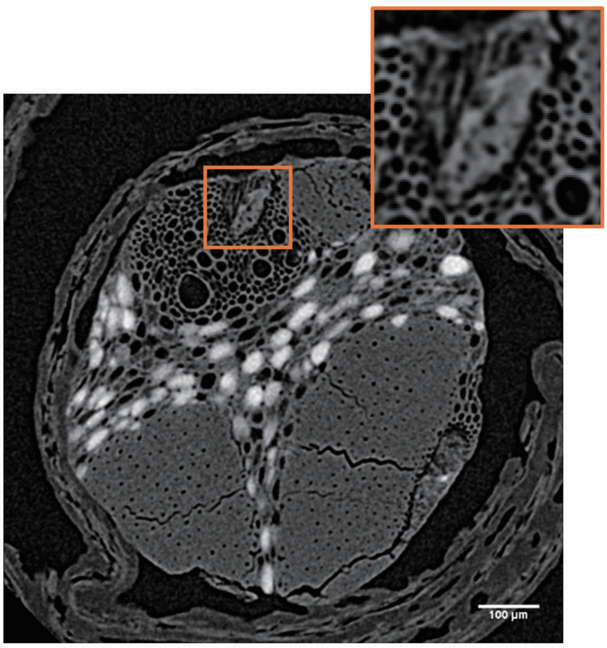
Projected image spatial resolution: 0.5 μm (no sharpening processing)

High resolution (3D)

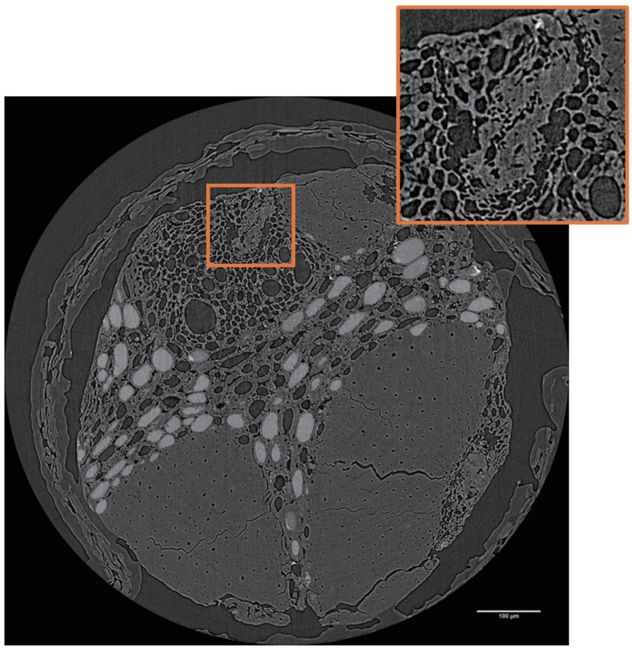




Bamboo cross-sectional image (device comparison)



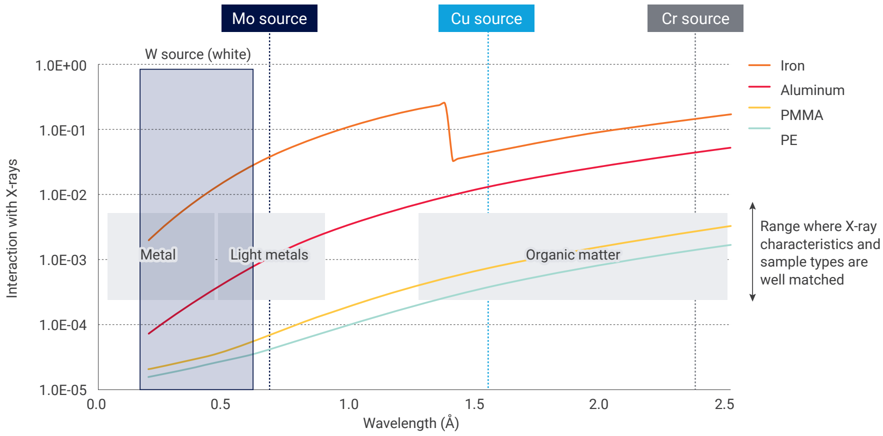
Conventional microCT system



nano3DX

High contrast

- Along with the conventional W source used for X-ray CT, the system offers selectable, dual source X-ray targets so you can match X-ray energy to the specimen and imaging goals (Cu/Mo standard; Cu/Cr and Cu/W optional)
- Cr and Cu provide lower-energy, longer-wavelength characteristic X-rays, well suited for low-density materials such as organic specimens
- Mo and W produce higher-energy, shorter-wavelength characteristic X-rays, making them effective for inorganic materials and light metals
- Using characteristic X-rays from Cr, Cu, or Mo, together with phase retrieval, enhances contrast for samples with subtle density differences

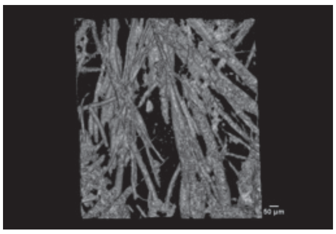


Representative sample images taken with each radiation source

W source

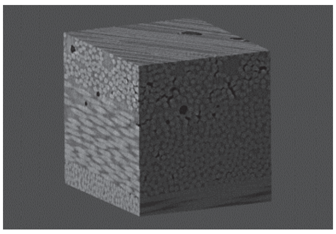
Electronic components

Mo source



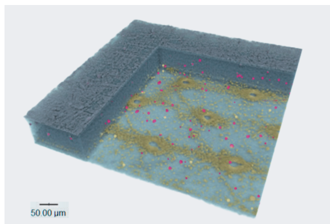
Al–C composite

Cu source



Carbon fiber reinforced resin

Cr source

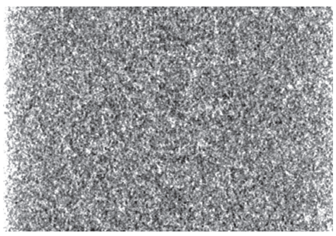


Multilayer organic film

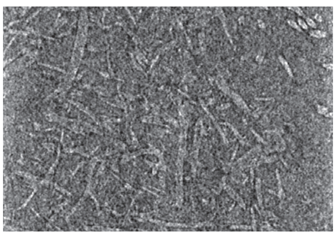
Examples of nonwoven fabric (organic material)

For nonwoven fabrics and other organic materials, longer-wavelength (lower-energy) sources such as Cr or Cu typically provide better contrast.

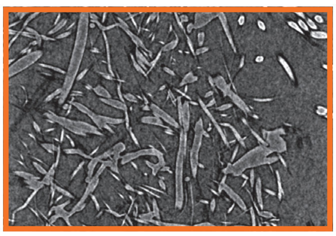
W source



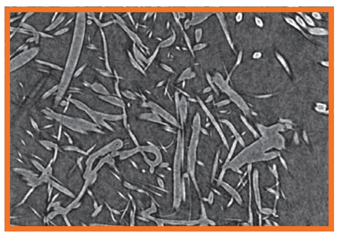
Mo source



Cu source



Cr source



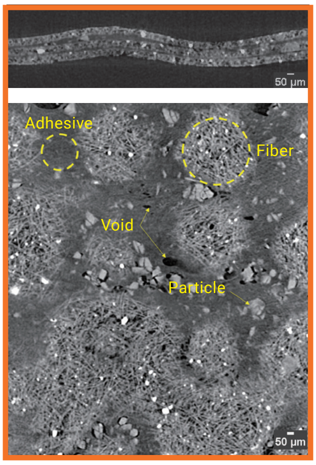
Nonwoven fabric (organic material)

Clearer contrast is typically achieved with longer-wavelength (lower-energy) characteristic X-rays, such as those from Cr or Cu. For organic materials like adhesive layers, Cu is often an effective choice for improving contrast.

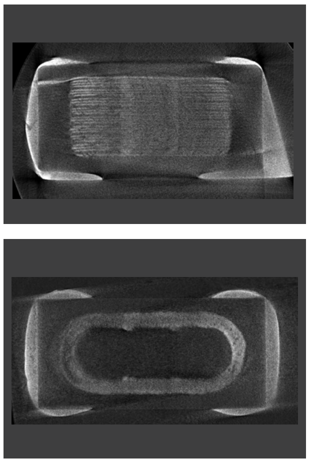
Example of photographing electronic components (metal)

For metals, a shorter wavelength W source provides clearer contrast.

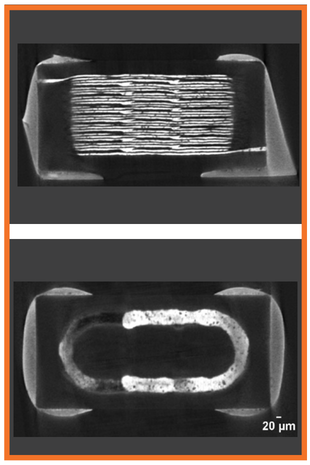
W source



Cu source



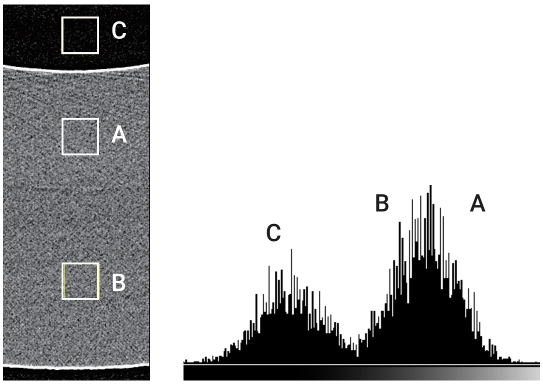
Mo source



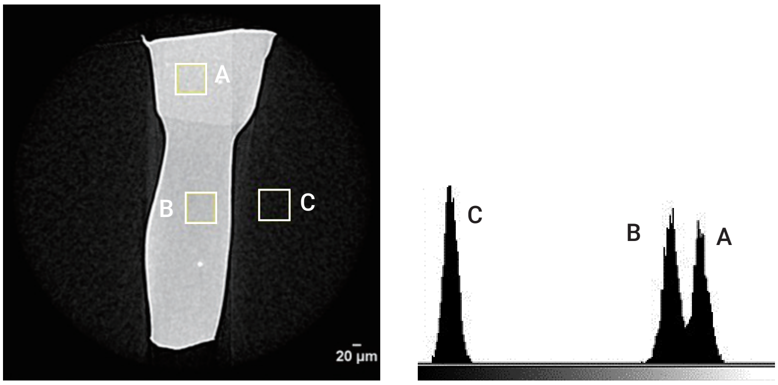
W source

Example of an adhesive layer (organic matter)

Characteristic X-rays, combined with phase retrieval, can maximize contrast even when density differences are small. For example, using the long-wavelength Cr source enables differentiation between HDPE (0.93 g/cm³) and LDPE (0.88 g/cm³).



Micro X-ray CT



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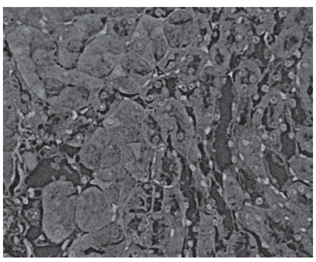
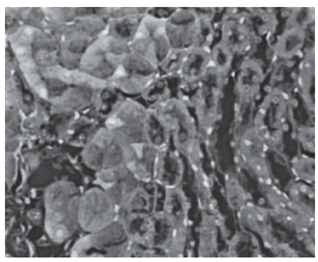
High-contrast imaging of biological soft tissues using phase retrieval

Even for biological soft tissues, where absorption contrast is often limited, phase retrieval enables high-contrast imaging.

※ Phase retrieval software is optional



Kidney, Cu source, no phase retrieval



Kidney, Cu source, with phase retrieval

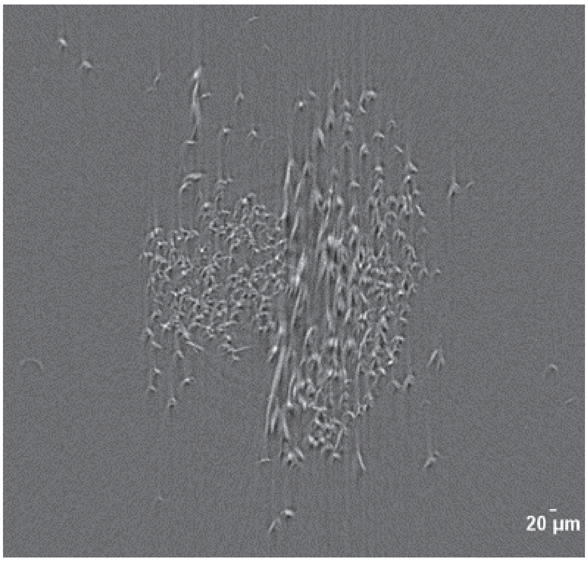
Sample provided by: Dr. Kengo Furuichi, Kanazawa Medical University

High speed in situ imaging

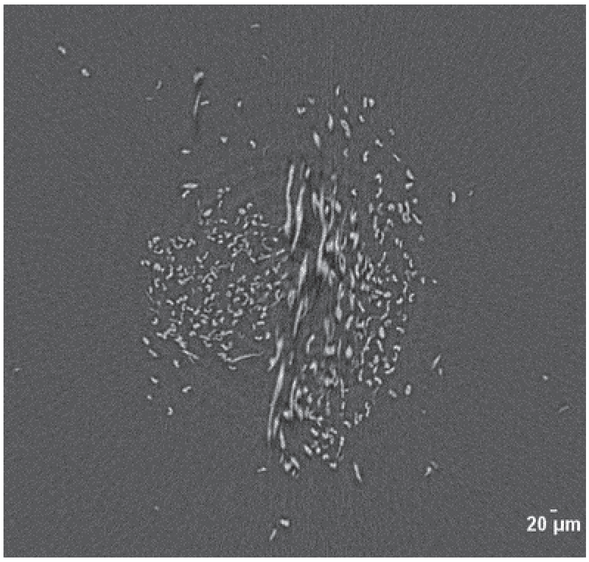
- The rotating anode X-ray source delivers roughly 100 X higher intensity than typical microfocus sources used for high-resolution imaging
- Continuous scanning with a high-speed X-ray detector
- Time-resolved studies using high-speed 2D radiography and CT acquisition modes

High-speed imaging of fibers

High-speed imaging reduces motion artifacts caused by sample deformation.



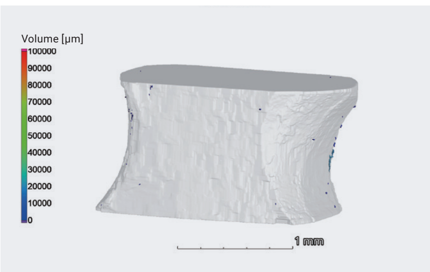
129 minute scan



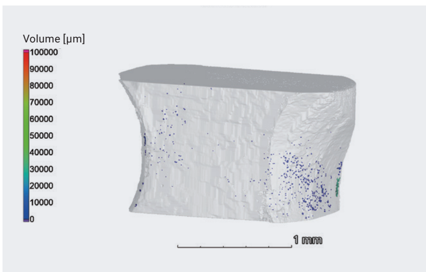
12-minute scan

In-situ imaging of carbon fiber reinforced resin

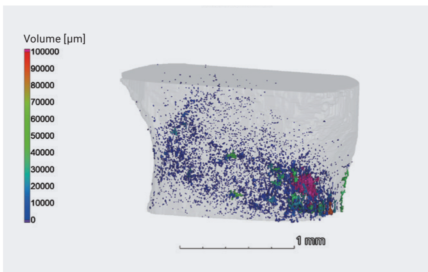
Using a tension attachment, in-situ imaging was performed. This allows for three-dimensional observation of changes over time in delamination and cracks inside the carbon fiber reinforced resin.



0 N



160 N

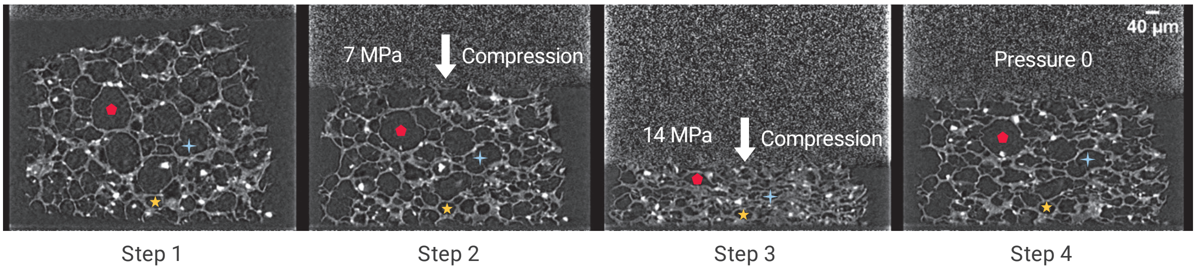


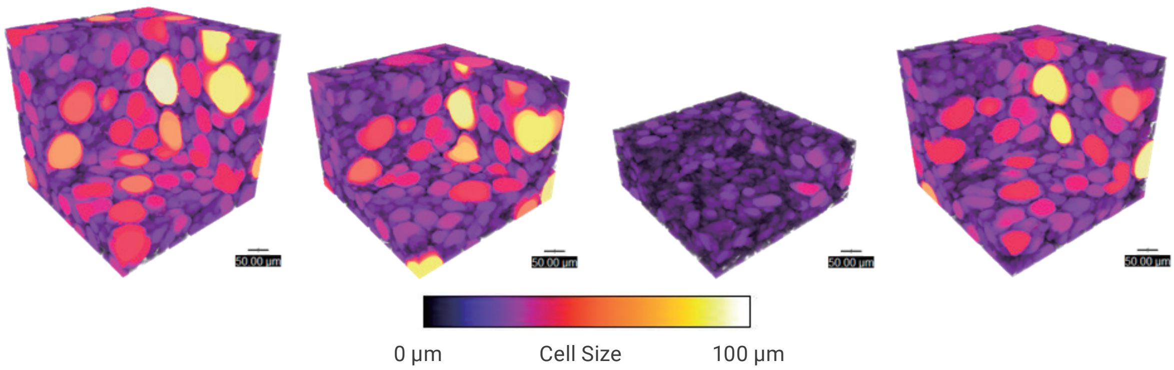
185 N

In situ image stages

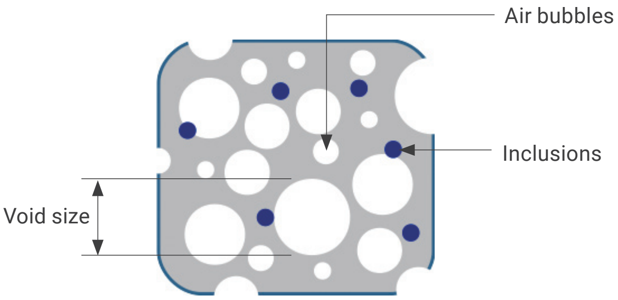
Heating	Cooling	Heating & Compression	Tensile
Temperature range: RT to 200 °C Sample size: 5 mmφ x 5 mm or less	Temperature range: -20 °C to RT Sample size: 8 mmφ x 4 mm or less	Compression range: 1 to 200 N (in 1 N increments) Temperature range: RT to 200 ℃ Sample size: 10 mmφ × 2 mm	Tensile range: 1 to 200 N (in 1 N increments) Sample size: Length 18 mm x Width 8 mm x Thickness 1.5 mm

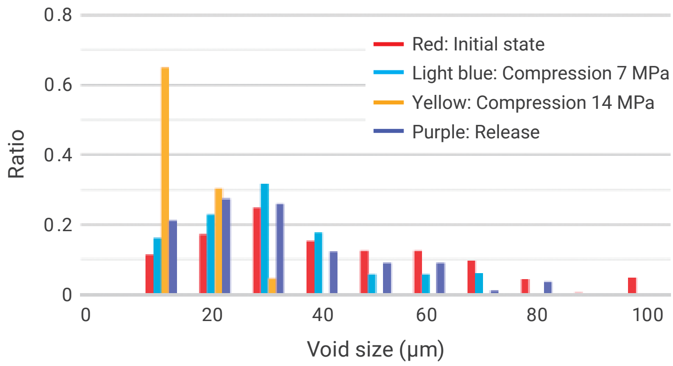
Example of foam resin compression (void volume analysis)





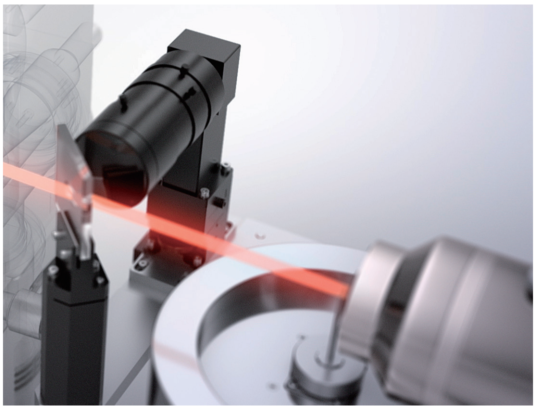
The voids decreased in size during compression and remained smaller afterward, without returning to their original dimensions.



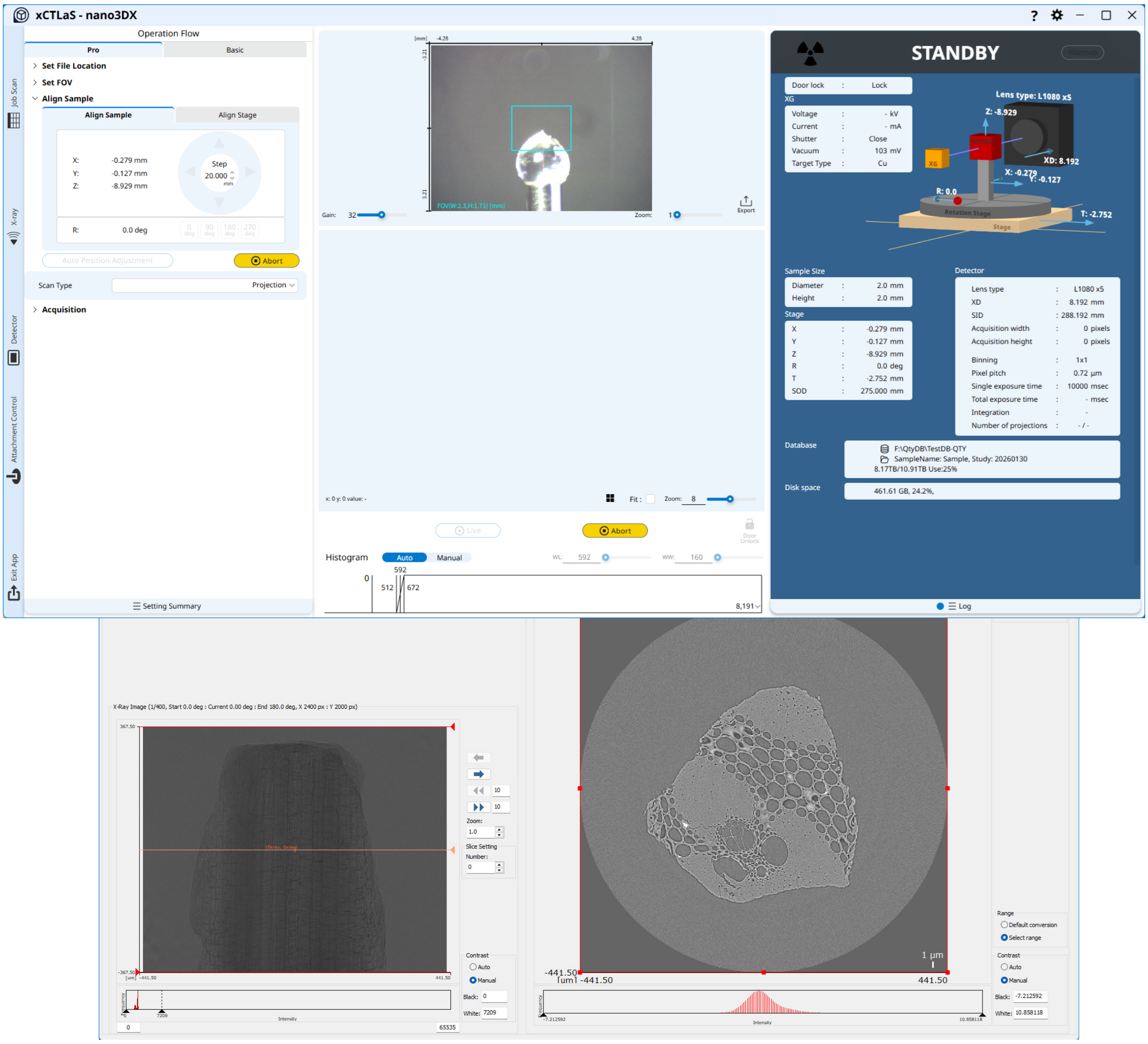


Easy to use software for CT imaging and reconstruction

- Select the X-ray target that best fits your sample from a dropdown list
- An on-axis optical camera lets you view the sample from the perspective of the X-ray beam
- Fast reconstruction software with built-in correction tools
- Automated workflows for CT scanning and reconstruction



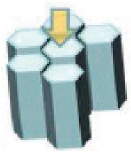
Control software



Large field of view projection image

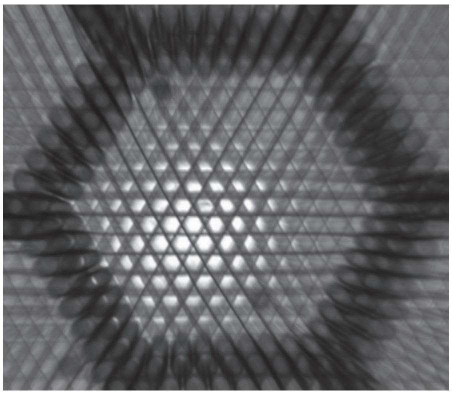
- Parallel beam geometry for low-distortion, true-to-geometry projections
- Translation stage travel: ±10 mm (horizontal), 40 mm (vertical)
- Built-in software for projection image stitching to create wide-field projection images

Distortion-free projected image



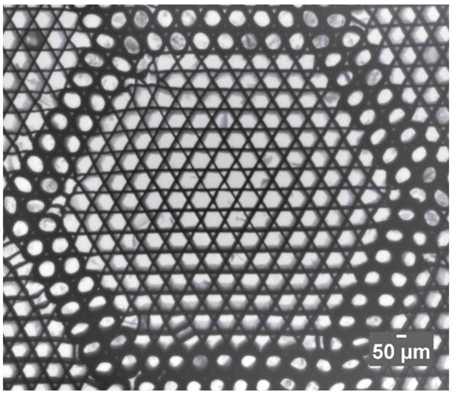
Polycapillary

Conventional cone beam microCT



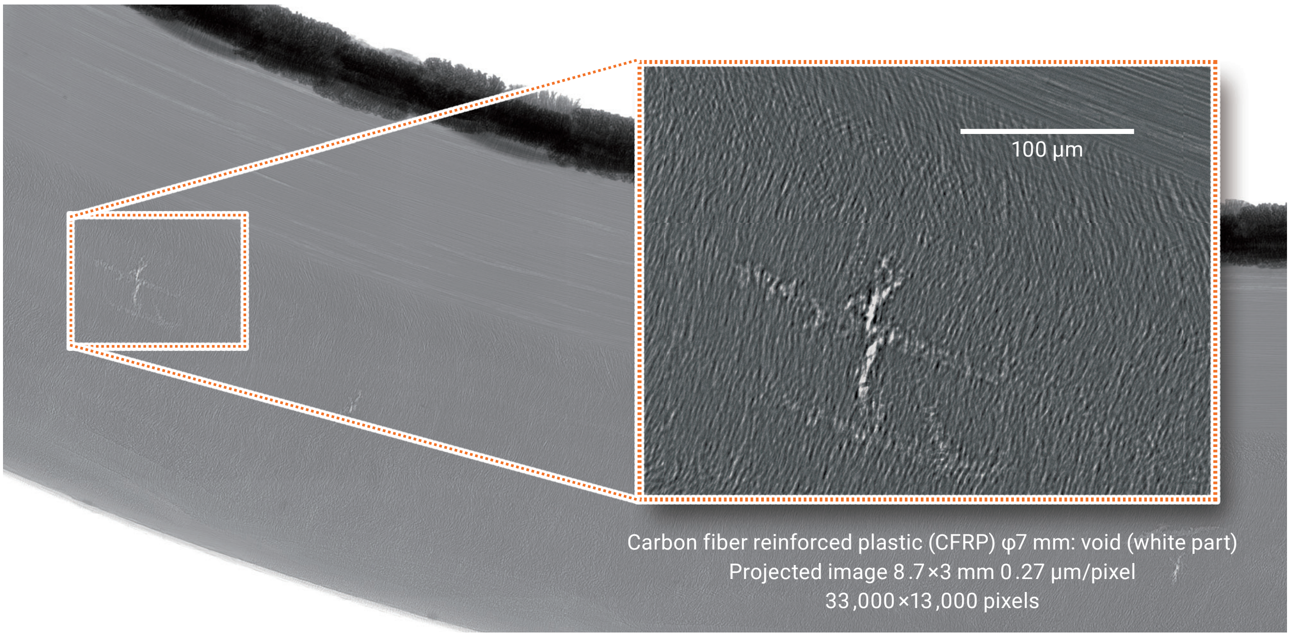
Projection images are distorted in conventional cone beam microCT systems, making structures appear different from their true geometry. CT scanning is required to view the true 3D shape.

nano3DX (parallel beam geometry)



With its parallel beam geometry, the nano3DX produces high-resolution, low-distortion projection images. This enables true-to-geometry shape observation with short exposure times, typically just a few seconds.

Samples can be imaged over a wide field of view while maintaining high resolution.

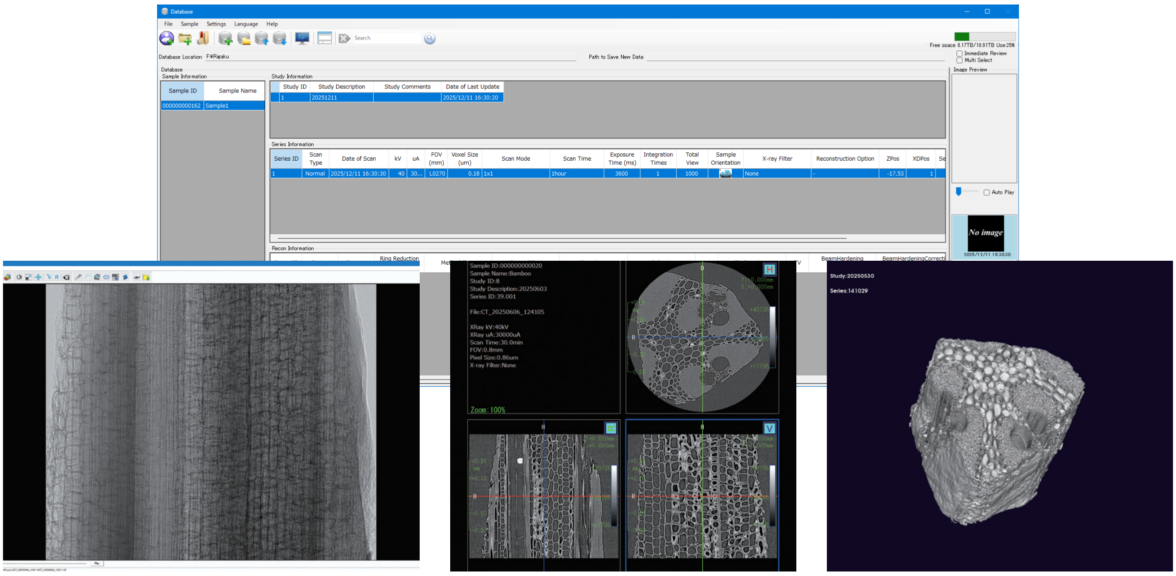


Carbon fiber reinforced plastic (CFRP) φ7 mm: void (white part)
Projected image 8.7×3 mm 0.27 µm/pixel
33,000×13,000 pixels

Comprehensive software for a wide range of applications

Acquired image data is managed in a database that centralizes display, organization, and export to a variety of user-selectable formats. The standard viewer supports 2D and 3D visualization and analysis.

Database software



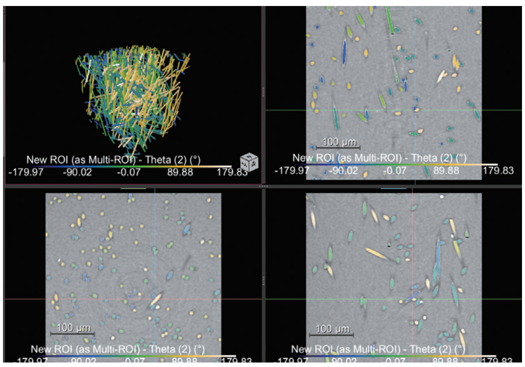
2D viewer software — Viewer software — 3D viewer software

External third-party packages read the native nano3DX 3D images to allow for segmentation and quantitative analysis.

Comet Technologies Canada Inc.

Dragonfly

Deep learning-based segmentation tools can analyze voids, particle distribution and size, fiber orientation, etc.



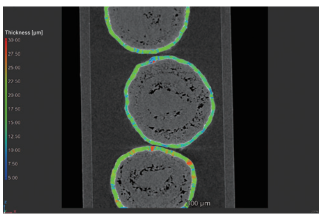
Dragonfly and the Dragonfly logos are trademarks of Comet Technologies Canada Inc.

Hexagon Manufacturing Intelligence

VGSTUDIO MAX

Using various modules, pores, particles, fiber orientation, sample shape, etc. can be analyzed.

Tomographic images software



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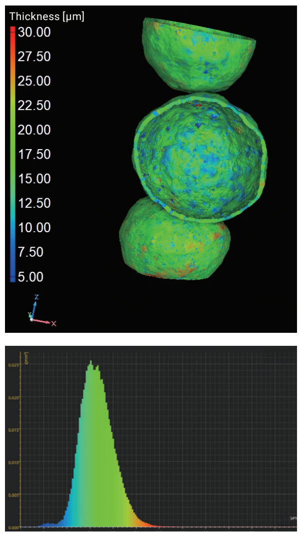
3D images



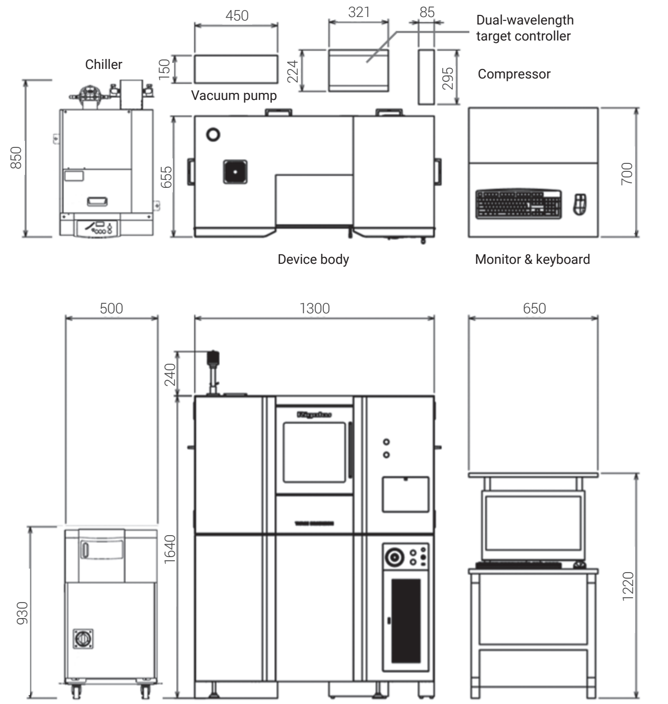
Specifications

Overall performance	
X-ray source	Rotating anode, high brightness X-ray generator
Tube voltage, tube current	~60 kV, ~30 mA
Target	Dual: Cu/Mo (option: Cu/Cr, Cu/W, Cr/Mo)
Detector	X-ray detector (CMOS)
Effective pixels	4800×4000 pixel
Voxel size	0.188~3.00 μm/voxel
Field of view	φ0.90×0.75~φ14.4×12.0 mm
Dynamic range	16 bit
Sample stage	Automatic 5-axis stage

Computer	
CPU	Intel Xeon
GPU	High performance
Memory	512 GB or more
Storage	16 TB or more
OS	Windows 11 64 bit
Monitor	27 inches

Software	
Control and reconfiguration software	X-ray CT imaging X-ray CT data reconstruction (Phase recovery function: optional)
Image data management Image display software	Database software: Organizes acquired datasets and metadata, and supports file conversion/export Image display software: Displays CT slice images and 3D volumes, with basic analysis tools
Optional Software	Reconstruction and image display software: Tomoshop (Midorino Research) Image display and analysis software: Dragonfly (Comet Technologies Canada Inc.) VGSTUDIO MAX (Hexagon Manufacturing Intelligence)

Device	
Dimensions	1300(W)×655(D)×1640(H) mm
Weight	650 kg
Chiller	Water-cooled indoor type (standard)
External leakage dose	1 μSv/h or less



Weight: 650 kg

Unit: mm

Installation

Requirements	
Chiller	1 unit (water-cooled indoor type: standard) Water supply rate: 8.3 L/min at 25°C, 15.5 L/min at 34°C (minimum)
Required power supply	Main body: AC three-phase 200V 15A Control/PC section: AC100 V 15 A Chiller: AC three-phase 200 V 10 A

Note: Please request system specification for detailed installation requirements.

Link to nano3DX video

https://www.youtube.com/watch?v=HU2FT-gNIKw

https://www.youtube.com/watch?v=5wpnfV9y8BA

nano3DX

High-resolution 3D X-ray microscope

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* The performance figures listed in this catalog are the results of tests conducted by Rigaku Corporation. We cannot guarantee that similar results will always be obtained in other environments.
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Specifications and appearance are subject to change without notice.